

Recommended buys

The very best products you can buy right now spanning 10 categories including gaming mice, graphic and sound cards, and E-Readers

Killer rigs

Building a rig? Don't know what components to include? Fret not, we'll sort you out

To print or not to print?

Brace yourself! Agent goes printer shopping and reports his finding

Agent 001

agent001@digit.in

Quite literally, and this is no exaggeration, but last month my trusty old printer broke down after close to a decade of solid performance. It just slowly whined, sputtered erratically and went eerily silent. So much for goodbyes, eh?

What that left me to do was to find a replacement printer real quick. Although I have been following the advancement in printer technology just like any other, I hadn't really tried to go hammer and tongs on a printer simply because I never had any real need to address. Without wasting any time, I got my hands on three top-selling printers in the market, one each from Canon, Epson and HP.

Epson L455, HP Deskjet Ink Advantage 4535, Canon Pixma G2000 are all multi-function printers which offer print, scan and copying capabilities out of the box. All three printers are relatively new into the market and are ideal for home users.

While the Epson L455 comes with its ink tank technology – where the ink needed to print documents and photos sit in bottles – the HP Deskjet 4535 comes with traditional colour cartridges. The Canon seems to offer a bit of a mix of both HP and Epson sport, from what I could tell. Of all the three printers, HP had the most features – like email print, Wi-Fi print, touch screen dial.



All three printers were relatively similar when it came to printing our 64 page DMystify. In terms of quality, there was a very close race between the HP and Canon – especially when it came to printing pictures on glossy and matte papers. The HP Deskjet 4535 churned out better quality high-resolution photos on A4 matte paper, while Canon enjoyed the upper hand when it came to printing the same photos on glossy sheets. The difference was distinctly apparent. The HP printer's ink consumption was highest of all three, but hey the results showed as to why.

It's all in the cost

I'll let you in on a secret of the printer industry. When it comes to ink-tank and laser printers, the total cost of ownership or TCO is the most important aspect that matters, forget everything else. This is where manufacturers make most money. Most manufacturers claim that their print rates are around ₹1 to ₹5 per page,

and that's true, but that rate is almost exclusively determined by the cost of the cartridge and its yield. What do you do when you have a choice between a ₹5000 printer that yields prints at ₹1 per page and one that costs ₹10000 and yields prints at ₹0.5 per page?

Here's what you do (considering a laser printer):

- Cost of printer: X
- Cost of cartridge: Y
- Page yield of cartridge: Z
- Your estimated print volume per year: A
- Estimated life of printer (years): 3 (going by the average warranty period, but you can choose a figure of your liking)
- Estimated print volume over the life of the printer: 3*A
- Number of cartridges required over the life cycle of the printer: (3*A)/Z
- Total cost of cartridges required:
 $B = Y * ((3*A)/Z)$

Therefore, the Cost of Ownership would be as follows:

$$COO = (X + B)/(3*A)$$

Everything else being equal, the lower this value, the better. For colour printers, you need to estimate the number of colour vs. black and white prints that you'd use and expand the formula to account for the cost of colour cartridges. Especially if you'll need CMYK cartridges. 

	Epson L455	HP Deskjet Ink Advantage 4535	Canon Pixma G2000
Cold Boot (sec)	35.99 sec	35.05 sec	24.77 sec
DMystify (sec)	683.11 sec	419.55 sec	539.17 sec
Photo (sec) (1 copy)	98.18 sec	140 sec	48.03 sec
Colour scan (300 DPI)	25 sec	20 sec	38 sec
Colour copy (sec)	38 sec	39 sec	71 sec